NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2023-2024

IE2104 – DIGITAL ELECTRONICS

April / May 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer all questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) In its simplest form, derive the product-of-sums (POS) logic expression for function $y(a, b, c, d)$ in the connections shown in Figure 1.

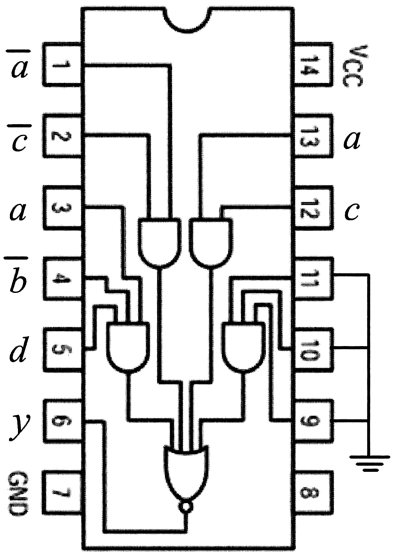


Figure 1

(8 Marks)

Note: Question No. 1 continues on page 2.

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(b) With minimum number of transistors, draw a CMOS circuit for the logic function, $y(a,b,c,d) = (\bar{a} \cdot c + a \cdot b) \cdot \bar{d} + \bar{b} \cdot \bar{c}$.

(10 Marks)

(c) In a base-14 number system, digits 10, 11, 12 and 13 are represented by A, B, C and D, respectively. Given the number $9CB_{14}$, state its decimal value and complete the counting sequence from $9CB_{14}$ by listing the next twenty base-14 numbers. How many digits are required for a base-14 number to represent decimal numbers up to 2 million?

(7 Marks)

2. (a) In the logic diagram shown in Figure 2, the 74LS86 exclusive-OR output drives several 74LS20 NAND inputs. The tied together 74LS20 inputs act like one LS input load in the LOW state and as separate LS input loads in the HIGH state. Thus, the 74LS86 output drives only five LS input loads in the LOW state and twelve LS input loads in the HIGH state. Using the information provided in Table 1 on page 3, determine if the fanout of 74LS86 has been exceeded, and explain your answer.

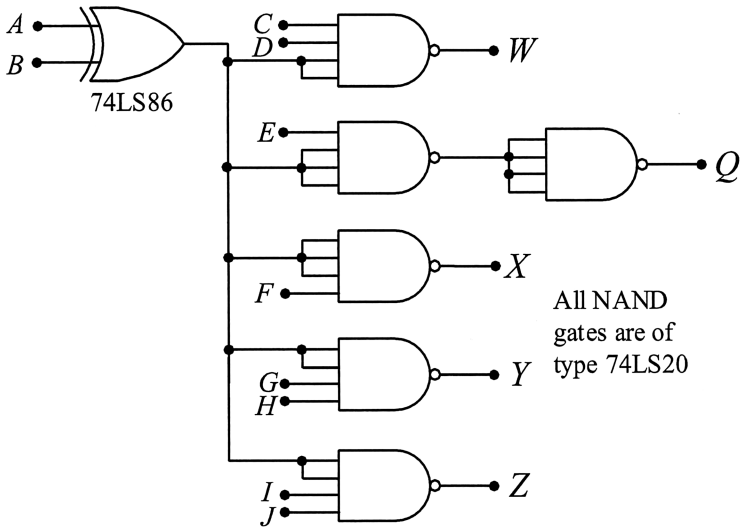


Figure 2

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Note: Question No. 2 continues on page 3.

Table 1: Operating range of TTL logic gates

TTL Logic Gate	Output HIGH Current I_{OHmax}	Output LOW Current I_{OLmax}	Input HIGH Current I_{IHmax}	Input LOW Current I_{ILmax}
74LS86 – Quad Exclusive-OR	–0.4 mA	8.0 mA	40 µA	–0.8 mA
74LS20 – Dual 4-Input NAND	–0.4 mA	8.0 mA	20 µA	–0.4 mA

(8 Marks)

(b) The function table of a 74151 8-to-1 multiplexer is given in Table 2. Design a circuit whose output is TRUE if and only if an odd number of its four inputs are TRUE. Use only one 74151 and one logic gate. Show your circuit diagram.

Table 2: Function table of 74151 8-to-1-multiplexer

Inputs				Outputs	
Select			Strobe		
C	B	A	$\overline{G}$	Y	W
x	x	x	1	0	1
0	0	0	0	D_0	$\overline{D_0}$
0	0	1	0	D_1	$\overline{D_1}$
0	1	0	0	D_2	$\overline{D_2}$
0	1	1	0	D_3	$\overline{D_3}$
1	0	0	0	D_4	$\overline{D_4}$
1	0	1	0	D_5	$\overline{D_5}$
1	1	0	0	D_6	$\overline{D_6}$
1	1	1	0	D_7	$\overline{D_7}$

(10 Marks)

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Note: Question No. 2 continues on page 4.

(c) The function table of a 74154 4-line-to-16-line decoder is given in Table 3. $G1$ and $G2$ are active low enable signals. Using only one 74154 and one logic gate, design a circuit to output a HIGH whenever the decimal value of the 4-bit binary input $DCBA$ is more than 5 but less than 9. Show your circuit diagram.

Table 3: Function table of 74154 4-line-to-16-line decoder

Inputs						Outputs														
$G1$	$G2$	D	C	B	A	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}	Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}
0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	1	0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1
0	0	0	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1
0	0	0	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1
0	0	1	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1
0	0	1	0	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1
0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1
0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1
0	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1
0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	0	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

(7 Marks)

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3. (a) Figure 3 shows a sequential circuit with input M and output Z. The function table of the 3-to-8 decoder is given in Table 4.

(i) Plot the K-maps for T_A, D_B and Z.
(ii) Tabulate the state-excitation table.
(iii) Draw the state diagram.

(13 Marks)

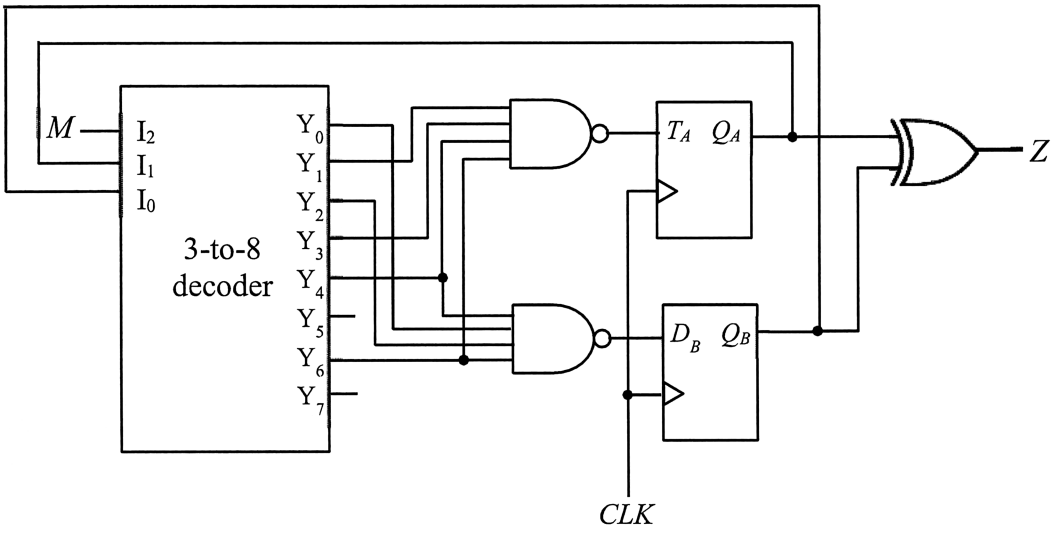


Figure 3

Table 4

Inputs			Outputs							
I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0	1
0	1	0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	1	1	0	1	1	1	1
1	0	1	1	1	0	1	1	1	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	1	1	1	1	1	1

Note: Question No. 3 continues on page 6.

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(b) The waveforms of the clock (*CLK*) and input (*M*) signals of the circuit in Figure 3 are given in Figure 4. The initial state of the finite state machine is $Q_A = 0$, and $Q_B = 0$. Assume there is no gate and wire delay. Copy Figure 4 onto your answer script and complete the waveforms of Q_A, Q_B and *Z*.

(7 Marks)

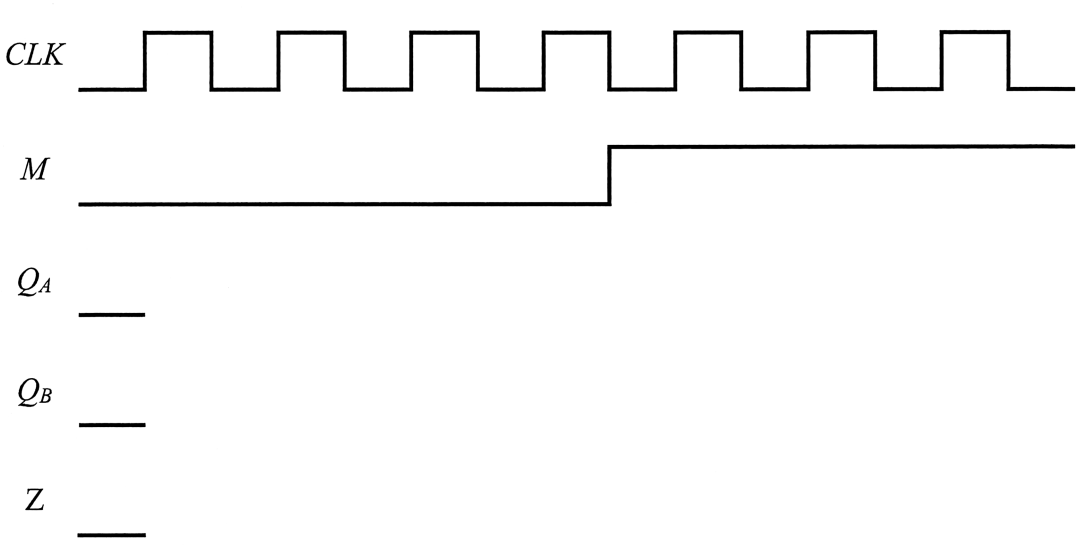


Figure 4

(c) Analyze and determine the function of this circuit.

(5 Marks)

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4. Figure 5 shows the circuit of a register.

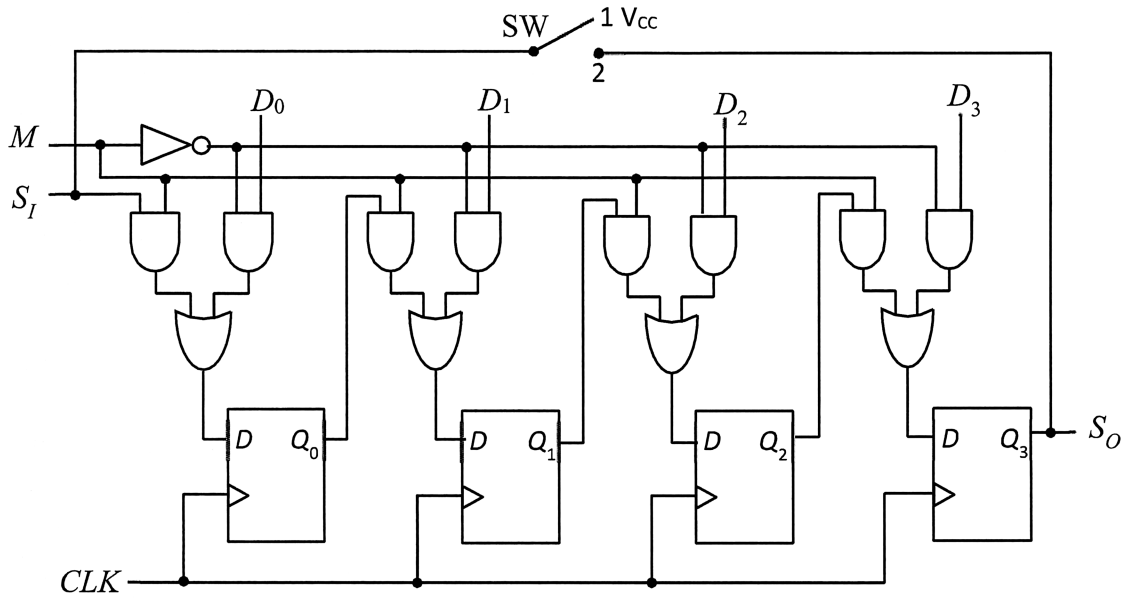


Figure 5

(a) Initially, $M = 0$, $D_0 = 0$, $D_1 = 0$, $D_2 = 1$, $D_3 = 0$ and the switch is at position 1. Then M changes from logic 0 to logic 1. The switch SW is kept at position 1 for the first two clock cycles after M changes to 1. On the falling edge of the second clock cycle, SW is switched to position 2 (connected) while M remains at logic 1. Determine the sequence of output bits generated at S_O for the first eight clock cycles when $M = 1$.

(8 Marks)

(b) Each two-input AND gate has a minimum propagation delay of 2 ns and a maximum propagation delay of 7 ns, and each two-input OR gate has a minimum propagation delay of 3 ns and a maximum propagation delay of 8 ns. Each D flip-flop has a setup time of 9 ns and a hold time of 6 ns. The minimum and maximum propagation delays of each D flip-flop are 4 ns and 12 ns, respectively. The frequency of the clock signal (*CLK*) is 25 MHz. If there is no clock skew, what is the setup time margin and hold time margin of this circuit when $M = 1$? What is the maximum clock frequency for this circuit to operate without timing violation?

(12 Marks)

(c) How much clock skew can the circuit tolerate before it experiences a hold time violation?

(5 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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